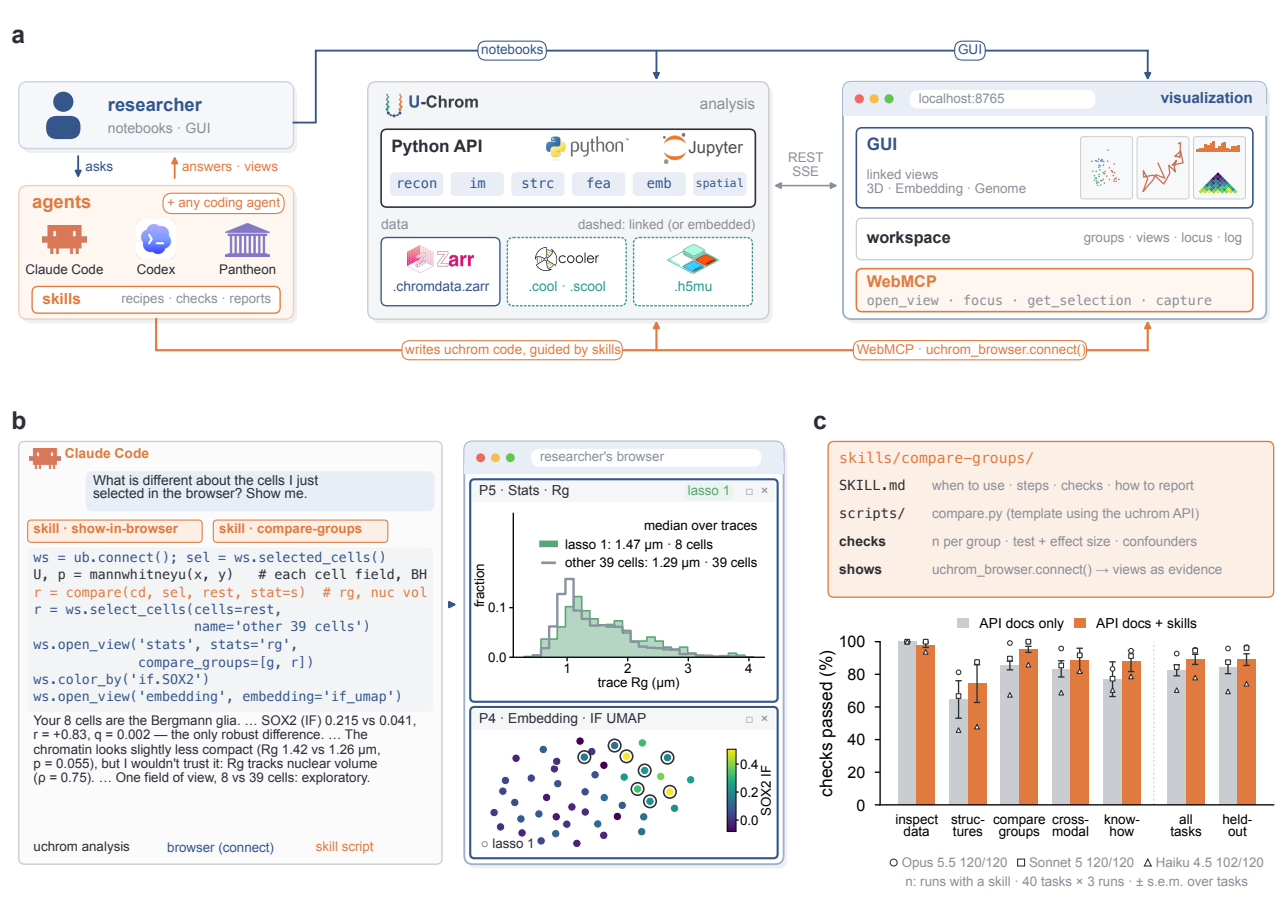

a
notebooks
GUI
researcher
notebooks · GUI
asks
answers · views
agents
+ any coding agent
Claude Code
Codex
Pantheon
skills
recipes · checks · reports
U-Chrom
analysis
Python API
python
Jupyter
recon
im
strc
fea
emb
spatial
data
dashed: linked (or embedded)
Zarr
.chromdata.zarr
cooler
.cool · .scool
.h5mu
REST
SSE
localhost:8765
visualization
GUI
linked views
3D · Embedding · Genome
workspace
groups · views · locus · log
WebMCP
open_view · focus · get_selection · capture
writes uchrom code, guided by skills
WebMCP · uchrom_browser.connect()
b
Claude Code
What is different about the cells I just selected in the browser? Show me.
skill · show-in-browser
skill · compare-groups
ws = ub.connect(); sel = ws.selected_cells()
U, p = mannwhitneyu(x, y) # each cell field, BH
r = compare(cd, sel, rest, stat=s) # rg, nuc vol
r = ws.select_cells(cells=rest,
name='other 39 cells')
ws.open_view('stats', stats='rg',
compare_groups=[g, r])
ws.color_by('if.SOX2')
ws.open_view('embedding', embedding='if_umap')
Your 8 cells are the Bergmann glia. … SOX2 (IF) 0.215 vs 0.041, r = +0.83, q = 0.002 — the only robust difference. … The chromatin looks slightly less compact (Rg 1.42 vs 1.26 µm, p = 0.055), but I wouldn't trust it: Rg tracks nuclear volume (ρ = 0.75). … One field of view, 8 vs 39 cells: exploratory.
uchrom analysis
browser (connect)
skill script
researcher's browser
P5 · Stats · Rg
lasso 1
median over traces
lasso 1: 1.47 µm · 8 cells
other 39 cells: 1.29 µm · 39 cells
fraction
0.1
0.0
1
2
3
4
trace Rg (µm)
P4 · Embedding · IF UMAP
lasso 1
0.4
0.2
0.0
SOX2 IF
c
skills/compare-groups/
SKILL.md
when to use · steps · checks · how to report
scripts/
compare.py (template using the uchrom API)
checks
n per group · test + effect size · confounders
shows
uchrom_browser.connect() → views as evidence
API docs only
API docs + skills
checks passed (%)
100
80
60
40
20
0
inspect data
struc-tures
compare groups
cross-modal
know-how
all tasks
held-out
Opus 5.5 120/120
Sonnet 5 120/120
Haiku 4.5 102/120
n: runs with a skill · 40 tasks × 3 runs · ± s.e.m. over tasks